



Final Internship Report

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Internship Place: Department of Statistical Sciences, University of Padua
Internship Period: 09/03/2025 - 31/07/2025 **Supervisor:** Prof. Mariangela Guidolin

Work carried out: During the internship, following tasks were completed:

1. Studying Fintech, including Financial Econometrics, Derivative Pricing, Stochastic Modeling, and ML/DL for Finance.
2. Studying Reinforcement Learning (RL), from basic algorithms like Dynamic Programming, Monte-Carlo, Temporal Difference, to advanced ones like Policy Gradient, REINFORCE, Actor-Critic Models.
3. Implementing Q-Learning algorithm on AMZN (Amazon) stock to check if RL works on stock trading.
4. Reading papers about applying RL in portfolio management, and papers of state-of-the-art (SOTA) RL models; then summarizing their mathematical theory with consensus description and notations.
5. Collecting the last 10-year stock data for three portfolio indexes in three financial markets: DJIA 30 (US) index, TOPIX 30 index (Japan), and VN30 index (Vietnam). There was a lack of data for VN30 index due to many young companies just established in the last 5 years.
6. Inspecting collected data by visualizing sector distribution and stocks' value over time, then make some remarks about economy of each country.
7. Preprocessing data by computing technical indicators as additional features, including (Moving Average Convergence Divergence (MACD), Relative Strength Index (RSI), Commodity Channel Index (CCI), and Average Directional Index (ADX).
8. Implementing four SOTA Actor-Critic RL models for portfolio management with preprocessed data. Four models are: Advantage Actor-Critic (A2C), Proximal Policy Optimization (PPO), Deep Deterministic Policy Gradient (DDPG), and Twin Delayed DDPG (TD3)
9. Running experiments and measure models performance; Plotting visualizations of metrics, then comparing, making comments and explaining the results.
10. Collecting additional data on top 30 Vietnam stocks (not necessarily in VN30 index) and running experiments again to check if the result is consistent with the one for DJIA 30 and TOPIX 30 index.
11. Drawing conclusions on which model is optimal for portfolio management, and suggesting some directions for future work.

Critical issues: During the internship, no critical issues were observed and all main tasks were completed.

Date: 31/07/2025

Signature